PostgreSQL

Database Notes — Knowledge Base

1. Overview

PostgreSQL is a free, open-source, object-relational database management system (ORDBMS) known for standards compliance, extensibility, and reliability. It has been in active development for over 30 years and is widely used for both transactional (OLTP) and analytical workloads.

2. Key Features

- Full ACID compliance for reliable transactions.
- Rich data types: JSON/JSONB, arrays, hstore, UUID, geometric types, and custom types.
- Advanced indexing: B-tree, Hash, GiST, GIN, BRIN — supports full-text search and spatial data (via PostGIS).
- MVCC (Multi-Version Concurrency Control) for high-concurrency reads/writes without heavy locking.
- Extensibility — custom functions, extensions (e.g., pg_stat_statements, TimescaleDB, PostGIS).
- Strong support for window functions, CTEs (Common Table Expressions), and recursive queries.

3. When to Choose PostgreSQL

It's a strong default choice for applications needing complex queries, strong consistency, and structured relational data, while also flexibly handling semi-structured JSON data (JSONB) — reducing the need for a separate document store in many cases.

4. Basic SQL Example

```
CREATE TABLE users (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(150) UNIQUE,
    metadata JSONB,
    created_at TIMESTAMP DEFAULT now()
);

SELECT name, metadata->>'plan' AS plan
FROM users
WHERE metadata @> '{"active": true}';
```

5. Strengths vs Trade-offs

Strengths	Trade-offs

Rich SQL feature set, extensible	Vertical scaling is easier than horizontal
Strong consistency (ACID)	More tuning needed at very large scale
JSONB gives semi-structured flexibility	Write throughput lower than some NoSQL stores
Mature ecosystem & tooling	Sharding requires extensions/manual setup